



The AI scaffold and engagement spectrum as a novel UDL-aligned system for supporting students with dyslexia

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Abstract

Artificial Intelligence (AI) is rapidly changing higher education, offering new opportunities to support diverse learners, including students with dyslexia, while raising important questions about ethics and assessment. Existing work on Universal Design for Learning (UDL) and assistive technologies highlights persistent barriers in reading, writing and planning, but there is limited guidance on how generative AI (GenAI) can be integrated into academic work safely and accessibly. This study addresses that gap through a convergent parallel mixed-methods design with qualitative priority (QUAL + quan). Semi-structured interviews with specialist Student Support and Wellbeing Services (SSWS) were conducted alongside an online student survey to identify recurring challenges, current tool use and attitudes towards AI. The analysis reveals difficulties with academic writing, critical reading and time management, patchy and sometimes unsustainable adoption of assistive tools and AI, and uncertainty about acceptable AI use and its implications for academic integrity. In response, the paper proposes a dual-component, UDL-aligned system for responsible AI-supported assessment. The instructor-facing *AI Engagement Spectrum* helps academics set clear, assignment-specific rules for permissible AI use, grounded in constructive alignment. The student-facing *AI Scaffold* translates these rules into a structured workflow that guides students with dyslexia through reading, planning, drafting and revising with AI support while preserving authorship and critical engagement. The primary contribution is a transferable framework that links institutional policy, pedagogical intent and students' study practices. It offers SSWSs, lecturers and educational designers a practical starting point for implementing AI in ways that are accessible, transparent and focused on supporting students with dyslexia.

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Keywords Universal design for learning (UDL) · Dyslexia · Generative AI for assessment · Assistive technology · Academic writing · Student support and wellbeing service

1 Introduction

Dyslexia is a common specific learning difference, estimated to affect around 5–10% of the population worldwide (Snowling, 2019), with significant implications for university-level study. Students with dyslexia often experience persistent difficulties in reading, spelling and written expression, alongside working-memory and processing-load constraints that affect planning and organisation (NHS, 2022; Richardson & Wydell, 2003; Swanson, 2020a). In practice, this can make it harder to keep up with reading lists, extract key points from dense texts and produce extended written assignments, contributing to attainment gaps and higher attrition risks compared to their peers (Richardson & Wydell, 2003). These challenges are amplified in digital learning environments, where most materials are delivered via screens and students are expected to juggle multiple online sources and platforms.

Universities commonly respond through a mix of individual accommodations and broader commitments to Universal Design for Learning (UDL). UDL advocates designing curricula that offer multiple means of engagement, representation and action/expression from the outset so that variability in learners' profiles, including dyslexia, is anticipated rather than treated as an exception (CAST, 2024; Rose & Meyer, 2002). For students with dyslexia, this can mean routine access to alternative representations (for example, Text-to-Speech (TTS), visual organisers and chunked instructions) and flexible ways of demonstrating understanding (Bäck et al., 2024; Rao, 2020). However, putting UDL into practice consistently remains challenging (Fovet, 2020; Rao, 2020), and there is limited guidance on how to translate UDL principles into day-to-day study workflows and assessment practices for specific groups such as students with dyslexia. Recent work has begun to explore UDL-informed approaches to AI curricula and inclusive AI education more broadly (Saborio-Taylor & Rojas-Ramírez, 2024; Song et al., 2024; Zhao et al., 2025b), but a concrete, operational framework for combining UDL with AI support in university contexts is still missing. The rapid emergence of AI-powered study tools extends this assistive technology landscape, but institutions currently offer limited concrete, UDL-aligned guidance on how such tools should be integrated into everyday study workflows for students with dyslexia.

Alongside traditional assistive technologies such as TTS, dictation, grammar checkers and mind-mapping tools (Lerga et al., 2021; Paudel & Acharya, 2024; Zallio & Ohashi, 2022), the rapid emergence of Generative Artificial Intelligence (GenAI) systems has created new possibilities for supporting students with dyslexia. Tools such as ChatGPT can summarise readings, suggest outlines, propose alternative phrasings and offer near-instant feedback on drafts (Holmes et al., 2019; Ji et al., 2023; Kasneci et al., 2023). Used carefully, these capabilities can reduce the time and effort spent on decoding dense text or overcoming “blank page” paralysis, allowing students to focus more on comprehension, idea generation and argumentation

(Farhah et al., 2025; Snowling, 2019; Zhao et al., 2025b). Early studies report that students perceive AI chatbots as helpful for brainstorming, clarification and planning, and many express positive views about their usefulness (Chan & Hu, 2023; Mahapatra, 2024).

At the same time, AI-supported study raises familiar concerns about academic integrity, cognitive offloading and equity. Without clear guidance, students may use GenAI to produce large portions of assessed work, undermining the validity of grades and obscuring what they themselves can do (Chaudhary et al., 2024; Risko & Gilbert, 2016; Vanderbilt University Center for Teaching, 2023). There is also the risk that learners over-delegate intellectual work to AI and accept fluent but incorrect outputs with minimal scrutiny (Ji et al., 2023; Kasneci et al., 2023). Institutional responses have begun to move from simple prohibition towards more nuanced policies that specify, at task level, which forms of AI use are acceptable, require disclosure of AI assistance and place responsibility for verification on students (American Psychological Association, 2023; Cornell University Center for Teaching Innovation, 2023). Yet these policies often remain high level; they rarely provide concrete, dyslexia-aware study workflows that help students use AI as assistive intelligence rather than as a shortcut.

These tensions are particularly acute for students with dyslexia. On the one hand, they stand to benefit from AI tools that can reduce extraneous reading and writing barriers and support planning and organisation. On the other, they may be especially vulnerable to using AI in ways that bypass core learning processes or expose them to integrity risks. What is needed is not simply more tools, but a structured, UDL-informed and constructively aligned system that connects institutional AI policies, module-level decisions and students' day-to-day study habits. In this paper we address that need by working with a university Student Support and Wellbeing Service (SSWS) and students to develop a dual-component pedagogical system, consisting of an instructor-facing *AI Engagement Spectrum* and a student-facing *AI Scaffold*, nested within a broader conceptual model for AI-supported assessment and study for students with dyslexia.

1.1 Aim and research questions

In light of these challenges and opportunities, this study set out to develop a practical framework for integrating AI-powered tools into the academic workflows of students with dyslexia in an ethical and effective manner. The goal was not only to understand the current pain points and perceptions around AI, but to produce a concrete solution in the form of a step-by-step workflow that guides students in using AI to support their reading, writing, and planning tasks while upholding academic integrity. The research was guided by three primary questions:

1. **RQ1.** What academic challenges do students with dyslexia face in reading, writing and planning their work?
2. **RQ2.** How do students and Student Support and Wellbeing Services (SSWS) staff perceive the role of AI tools in these tasks, and what concerns arise (for example around academic integrity)?

3. **RQ3.** How can a dual-component pedagogical system be designed to guide both students and instructors in the ethical and effective use of AI? Specifically:
 1. How can an instructor-facing model, the *AI Engagement Spectrum*, grounded in the principles of constructive alignment, enable academics to set clear guidelines for AI use?
 2. How can a structured, UDL-aligned workflow, the *AI Scaffold* be designed to support students with dyslexia in their academic tasks while mitigating risks such as uncritical cognitive offloading?

By answering these questions, the study aims to bridge the gap between the unmet needs of students with dyslexia and the cautious adoption of new AI tools, culminating in a framework that universities can adopt to support inclusive and responsible learning.

1.2 Contributions

We first carried out an exploratory needs analysis of students and staff (Contribution 1). In response, we developed a transferable conceptual model (Contribution 2) that is realised as a practical, dual-component system and grounded in an integrated application of UDL and constructive alignment (Contribution 3). The contributions are primarily conceptual and design-oriented: the empirical work provides a grounded basis for the framework rather than a summative evaluation of its impact.

1. An exploratory needs analysis, grounded in mixed-methods data from students and specialist staff, that identifies the specific academic challenges and AI-related uncertainties the *AI Scaffold* framework is designed to address.
2. A transferable conceptual framework, the Nested Scaffolding Model, which offers a multi-level architecture (institutional, pedagogical, procedural and cognitive) for responsible AI integration. This paper details the two central layers of this model:
 1. The Pedagogical Scaffold: an instructor-facing model, the *AI Engagement Spectrum*, grounded in constructive alignment, that empowers academics to define clear, assignment-specific rules for permissible AI use.
 2. The Procedural Scaffold: a student-facing workflow, the *AI Scaffold*, that provides a UDL-aligned, step-by-step process for using AI ethically and effectively within the boundaries set by the instructor.
3. An integrated application of UDL and constructive alignment that shows how these two approaches can be combined to design AI-inclusive pedagogy which supports diverse learners, including students with dyslexia, while preserving academic rigour.

2 Literature review

This review synthesises research on dyslexia, assistive technologies, GenAI and assessment to motivate a UDL-aligned AI workflow for students with dyslexia. It outlines key academic challenges and existing supports, examines GenAI's affordances and risks (including cognitive offloading and integrity concerns), and positions constructive alignment and UDL as the pedagogical lenses underpinning the AI Engagement Spectrum and AI Scaffold framework.

2.1 Dyslexia in higher education

Students with dyslexia typically encounter pronounced difficulties with text-heavy materials. Reading scholarly articles or lengthy textbooks can be slow and exhausting, often compromising comprehension, while extended writing may involve problems with planning, organisation, and coherence (Mortimore & Crozier, 2006; Snowling & Hulme, 2012). These challenges are associated with attainment gaps and higher attrition rates in higher education (Pino & Mortari, 2014; Richardson & Wydell, 2003).

Difficulties can intensify in digital learning environments where most content is screen-based. Eye-movement and screen-reading studies report more and longer fixations and greater cognitive effort in dyslexic readers, highlighting the need for accessible presentation (Prado et al., 2007; Rello & Baeza-Yates, 2017; Le Cunff et al., 2024). Managing multiple online sources also taxes attention and working memory, which are often constrained in dyslexic learners (Swanson, 2020b; Le Cunff et al., 2024). Support such as read-aloud technology can improve comprehension under high load, reinforcing the case for accessible, multimodal materials (Schiavo et al., 2021).

Alongside these challenges, some studies suggest specific visual–spatial advantages on global processing tasks, while other work finds no general superiority, indicating any strengths are likely task-specific (von Károlyi et al., 2003; Winner et al., 2001). Effective support should therefore leverage potential idea-generation/holistic thinking strengths while mitigating barriers in text-intensive tasks (MacCullagh et al., 2017; Mortimore & Crozier, 2006; Snowling & Hulme, 2012).

2.2 Existing assistive tools and their limitations

Universities commonly provide text-to-speech, speech-to-text, word prediction, concept-mapping tools, and screen readers; students report better access, reduced fatigue, and smoother drafting when these are part of routine workflows (Bäck et al., 2024; Pino & Mortari, 2014). Concept mapping and related visual organisers can aid organisation and understanding across domains, supporting planning demands common in extended writing (Nesbit & Adesope, 2006). Beyond basic utilities, multimodal supports that combine visual and auditory channels can improve comprehension under high load, especially when embedded within tasks rather than bolted on Mayer (2009); Schiavo et al. (2021).

However, support is often fragmented: tools sit outside core virtual learning environments, and guidance typically addresses decoding or transcription rather than argumentation, critical reading, or synthesis (Pino & Mortari, 2014; Seale, 2013). Even when advanced tools are available, students frequently lack strategic instruction on when and how to use them across the assignment lifecycle (Pino & Mortari, 2014). Access and uptake vary with diagnosis, funding, and prior assessment, leaving some groups, such as late-diagnosed or international students, less likely to receive formal support (Pino & Mortari, 2014). As a result, current tools help with sub-tasks but rarely provide a coherent, pedagogy-driven pathway through complex assignments, underscoring the need for alignment between tools and intended learning outcomes (Biggs, 1996; Seale, 2013).

2.3 GenAI as a new support modality

The emergence of GenAI tools has introduced new ways to support learners, including those with dyslexia. Large language models can summarise texts, generate explanations, suggest structures, and provide examples in accessible language (Kasneci et al., 2023; Weng et al., 2024). For students who struggle with decoding dense prose, GenAI can act as a bridge: it can rephrase material, generate outlines, and offer step-by-step prompts that sustain engagement with challenging tasks (Snowling & Hulme, 2012; Swanson, 2020b; Zhao et al., 2025a). GenAI can also provide rapid feedback on draft text, enabling iterative refinement and experimentation with expression (Chan & Hu, 2023; Weng et al., 2024).

At the same time, GenAI differs from traditional assistive tools. Rather than compensating only for decoding or transcription bottlenecks, it can generate new content (arguments, examples, extended prose), which blurs the boundary between support and substitution (Kirschner et al., 2006). Without clear guidance, students may bypass core cognitive work; moreover, LLMs can produce fluent but incorrect or fabricated content, requiring verification to avoid error propagation (Buçinca et al., 2021; Ji et al., 2023).

2.4 The challenge of cognitive offloading in AI-supported learning

Cognitive offloading refers to the use of external tools or resources to reduce burdens on internal cognitive processes, such as working memory or attention (Clark & Chalmers, 1998; Risko & Gilbert, 2016). In education, well-designed offloading can free up resources for higher-order thinking; for example, diagrams or checklists can support planning so learners can focus on argumentation rather than logistics (Larkin & Simon, 1987; Mayer, 2009). However, offloading becomes problematic when it displaces essential cognitive work and thereby undermines learning (Kirschner et al., 2006; Risko & Gilbert, 2016).

GenAI intensifies these tensions. Tools that summarise readings, generate essay plans, or suggest paragraphs may shift idea generation and coherence management to external systems (Kasneci et al., 2023). When AI routinely produces arguments, examples, or full sections of text, students may engage less deeply with sources and develop weaker independent reading and writing skills, echoing concerns about

reduced learning under minimally guided instruction (Kirschner et al., 2006). Experimental and survey studies show productivity gains, while effects on underlying skills and long-term learning remain uncertain; some learners report becoming reliant on AI suggestions to start or complete tasks (Lee et al., 2025; Noy & Zhang, 2023; Weng et al., 2024). For students who already lack confidence in reading and writing, such as many with dyslexia, there is a risk that scaffolds slide into substitutions rather than gradually fading support (Aleven & Koedinger, 2002; Renkl & Atkinson, 2003).

Accuracy and trustworthiness present a parallel challenge: LLMs can produce fluent but incorrect or fabricated content, including non-existent references or distorted summaries (Ji et al., 2023). Over-trust without verification can encourage uncritical acceptance of AI-generated material and reduce active engagement with sources (Buçinca et al., 2021). Accordingly, responsible integration should specify which cognitive processes are supported and which must remain with the learner, and embed verification and reflection so offloading enhances rather than replaces independent critical thought (Aleven & Koedinger, 2002; Buçinca et al., 2021).

2.5 Student attitudes toward assistive technology and AI

Students' attitudes toward assistive and AI tools are often ambivalent: many value tangible benefits (access, reduced effort, smoother drafting) yet simultaneously report stigma or fear being perceived as "cheating" (McNicholl et al., 2021; Pino & Mortari, 2014). Among dyslexic students in particular, qualitative studies note that day-to-day experiences with support, training, and the visibility of tool use shape confidence and willingness to engage (Bäck et al., 2024; MacCullagh et al., 2017).

Similar ambivalence appears around GenAI: students describe usefulness for brainstorming, summarising, and feedback, but also worry about fairness, surveillance, and misconduct boundaries (Chan & Hu, 2023; Weng et al., 2024). Those who feel less confident in academic writing may be especially drawn to paraphrasing or drafting assistance while remaining uncertain about what counts as legitimate support (Chan & Hu, 2023).

Attitudes improve when expectations are explicit and support is integrated into mainstream provision rather than bolted on: clear, ILO-aligned guidance normalises appropriate use and reduces anxiety, whereas fragmented provision reinforces otherness (Pino & Mortari, 2014; Seale, 2013). Constructive alignment at the course level helps delineate where tools may scaffold access or expression and where students must demonstrate unaided cognitive work (Biggs, 1996).

2.6 Academic integrity and the shift towards pedagogical guidance

The rapid spread of GenAI prompted early bans and a focus on detection, but scholarship highlights enforcement limits and argues for guidance that clarifies how AI may be used for learning (Cotton et al., 2023; Eke, 2023). Recent reviews likewise emphasise redesigning assessment and providing disclosure rules over blanket prohibition (Weng et al., 2024).

This shift intersects with accessibility. Clear statements about permissible AI use can legitimise routine supports for students who rely on assistive/AI tools, whereas

generic policies risk ambiguity and stigma (Pino & Mortari, 2014; Seale, 2013). Students also report anxiety about fairness and misconduct boundaries, underscoring the need for explicit, course-level expectations (Chan & Hu, 2023).

Linking permissions to intended learning outcomes operationalises this guidance: constructive alignment specifies where tools may scaffold access or expression and where learners must demonstrate unaided cognitive work (Biggs, 1996; Cotton et al., 2023).

2.7 Constructive alignment as framework for AI pedagogy

Constructive alignment turns high-level AI policies into concrete assessment design by ensuring coherence between intended learning outcomes (ILOs), teaching–learning activities, and assessment tasks (Biggs, 1996; Biggs et al., 2023). Assessment is designed to elicit the cognitive evidence specified in the ILOs, rather than merely grading products.

Bloom’s taxonomy is commonly used to classify the targeted cognitive operations, from remembering and understanding through applying, analysing, evaluating, and creating (Anderson & Krathwohl, 2001; Bloom et al., 1956). In the GenAI era, this framing invites instructors to specify which aspects of task performance may be legitimately supported by tools and which must be demonstrated unaided by the student (Cotton et al., 2023; Weng et al., 2024).

Aligned designs can weave AI into learning by (a) generating practice items or exemplars, (b) asking students to critique or improve AI outputs, and (c) requiring documentation and justification of AI use as part of the submission (Baig & Yadegaridehkordi, 2024; Cotton et al., 2023; Kasneci et al., 2023). Studies also show that transparency about expectations and AI’s role improves perceived fairness and learning value (Chan & Hu, 2023; Luo, 2024). In short, applying constructive alignment here means treating AI use as part of the designed assessment process, not an afterthought or merely a regulated risk (Biggs et al., 2023).

2.8 UDL as a framework for AI pedagogy and its critiques

UDL builds flexibility into how information is presented, how students act on and express learning, and how they engage and sustain effort (Rose & Meyer, 2002). Reviews show UDL is used to plan curricula and technology supports across settings, though implementation varies and is often under-specified (Bray, 2024; Rao et al., 2014).

For students with dyslexia, UDL-informed design typically includes multimodal materials, explicit structure for complex tasks, and alternative modes of demonstrating understanding, reducing avoidable barriers without lowering expectations (Bäck et al., 2024; Pino & Mortari, 2014). When these supports are embedded in assessed tasks, they can help maintain access while preserving academic standards (Rao et al., 2014).

Critiques arise when UDL is interpreted as eliminating difficulty rather than unnecessary barriers. Scholars caution that evidence for UDL’s effectiveness is uneven and that misapplications risk echoing problems seen with “learning styles” claims (Boy-

sen, 2024; Edyburn, 2010). In higher education, a better balance is to pair UDL with desirable difficulties, conditions that feel effortful yet strengthen learning, so challenge is productive rather than punitive (Bjork, 1994; de Bruin, 2023).

In the GenAI context, this balance implies supports that lower extraneous load (e.g., simplification, structure, multimodal prompts) while requiring higher-order work and reflective decision-making about when and how to use AI (Kasneci et al., 2023). Accordingly, AI-supported assessments should be both UDL-informed and explicit about which cognitive processes must remain with the learner.

3 Methodology

This study used a convergent parallel mixed-methods design with qualitative priority (QUAL + quan) to develop and refine an AI integration framework for supporting students with dyslexia (Creswell & Plano Clark, 2017; Johnson & Onwuegbuzie, 2004). In a convergent parallel mixed methods design, qualitative and quantitative data are collected during the same study phase, analysed separately, and then merged for interpretation, and the notation QUAL + quan indicates qualitative priority in the integration. Semi-structured interviews with SSWS formed the lead strand, providing depth on challenges, existing practices and requirements from an expert perspective. Alongside this, a cross-sectional student survey provided descriptive breadth on students' experiences and attitudes to AI tools in academic work. Figure 1 summarises the mixed methods design, showing concurrent timing (QUAL + quan), separate analyses, the integration step, and the two subsequent validation activities. The two datasets were analysed separately and then integrated through comparison of themes and survey patterns to inform the design of a structured AI integration pedagogical system, followed by two layers of validation, review by SSWS and review against established pedagogical principles. Together, these strands addressed our research questions by characterising support needs and current tool use (RQ1–RQ2) and informing the co-design and refinement of the AI Engagement Spectrum and AI Scaffold framework (RQ3).

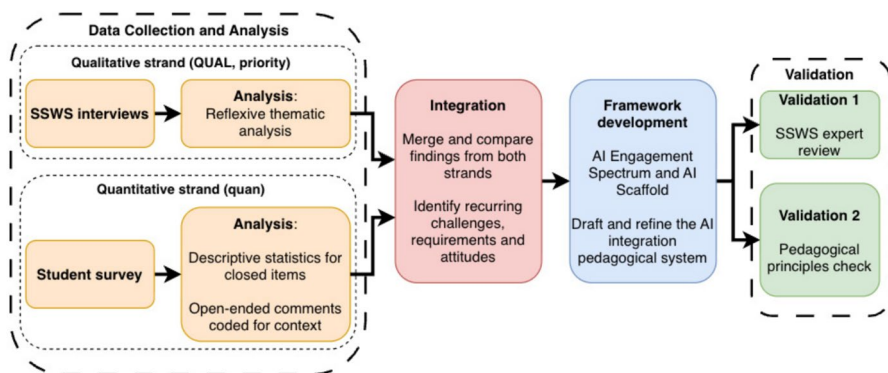


Fig. 1 Convergent parallel mixed methods with qualitative priority (QUAL + quan)

This user-centred, co-design-oriented approach is well suited to developing practical interventions in emerging technological areas where best practices are not yet established, and it aligns with wider work on inclusive design that foregrounds stakeholder voices in the creation of educational tools (CAST, 2024).

3.1 Data collection

3.1.1 Interviews with SSWS

We conducted one-on-one semi-structured interviews with three members of the SSWS at a UK university: two specialist study skills tutors, who regularly support students with dyslexia, and one assistive technology officer with expertise in integrating digital tools into study practices. Each interview lasted approximately 45–60 minutes and followed an interview guide structured around five broad topics: (1) common academic challenges for students with dyslexia, (2) existing support tools and strategies, (3) awareness of student use of AI tools, (4) concerns and institutional stances regarding AI, and (5) ideas for safe and effective integration of AI into study support. Questions were open-ended, with optional prompts to ensure sufficient depth on each topic and coverage of issues relevant to the research questions. The guide was developed specifically for this study, informed by the literature on dyslexia, assistive technologies and AI-supported study summarised in Section 2, and by prior discussions with SSWS about students' needs and institutional concerns. An outline of the interview guide, including example questions under each topic area, is provided in Appendix A. All interviews were conducted with informed consent, recorded and transcribed verbatim.

3.1.2 Student survey

Alongside the SSWS interviews, we designed an online survey to capture students' first-hand perspectives on using AI tools in academic work. The survey was distributed via SSWS and departmental mailing lists and shared through student social media groups. It remained open for two weeks. A total of 37 students responded, including undergraduate and postgraduate students from ten departments. Dyslexia was not a formal inclusion criterion. To protect privacy, participation did not require disability disclosure and the survey did not ask students to report diagnoses. This approach provided baseline perceptions and usage patterns of AI tools in the wider student population, while still gathering input from a subset of students with learning differences.

The survey combined multiple-choice, Likert-scale and open-ended items. Items were organised into four main sections: (1) study challenges (for example planning, reading, writing and managing workload), (2) current tool usage (including both assistive technologies and mainstream AI-powered tools), (3) perceptions of AI tools (perceived benefits, risks and impact on learning) and (4) ethical concerns and guidance (for example worries about integrity, plagiarism and clarity of university rules). All items were written specifically for this study, informed by the literature

on dyslexia, UDL and AI-supported study, and by consultation with SSWS. The full questionnaire is available in Appendix B.

3.2 Participants

The qualitative strand comprised three members of the university's SSWS, two specialist study skills tutors and one assistive technology officer. SSWS staff were selected because they work directly with students with dyslexia across disciplines and year groups, and are responsible for recommending and supporting the use of both traditional assistive technologies and newer AI-powered tools. In this institutional context they are the primary expert group for dyslexia-related study support, and were therefore well placed to comment on recurring challenges, current practices and requirements for any new workflow. Given the exploratory, design-oriented purpose of the qualitative strand, the aim was to obtain a focused, expert account rather than a demographically representative sample. The small, context-specific sample is treated as a limitation and we do not claim data saturation; instead, interviews were used to surface key patterns and practical considerations to inform the framework.

The quantitative strand comprised 37 students from ten departments across the university, spanning both undergraduate and postgraduate programmes. The survey included structured items that generated quantitative data (for example multiple-choice selections and Likert-type ratings) alongside open-ended prompts that generated qualitative comments for contextualisation. Recruitment was by open invitation via SSWS and departmental mailing lists, with no requirement to disclose a formal diagnosis. To avoid any pressure to reveal disability status, the survey did not ask students to report diagnoses directly, instead it included an optional item asking whether they drew on SSWS or similar disability-related study support. This design allowed us to situate the needs of students with dyslexia within a broader picture of AI use while respecting privacy. However, it also means that the precise number of formally diagnosed dyslexic respondents is unknown and the sample is small and self-selecting. Survey findings are therefore interpreted as exploratory and indicative rather than statistically generalisable, a limitation revisited in Section 6.5.

3.3 Data analysis

For the qualitative strand, the first author familiarised themselves with the data by reading transcripts and making initial notes, then generated inductive codes for barriers, existing supports and opportunities for AI-assisted study. Codes were collated into candidate themes, which were reviewed and refined with the wider research team, with particular attention to how they related to the three research questions (RQ1–RQ3) and to design decisions in the framework. All transcripts were anonymised, and reported findings draw on aggregated observations and anonymised quotations. This process followed principles of reflexive thematic analysis (Braun & Clarke, 2006). In line with reflexive thematic analysis, we treated codes as concise labels capturing features of the data relevant to our research questions (for example overload from reading lists, uncertainty about AI policy and time pressure in drafting), and themes as patterned meanings across the dataset (for instance reading as a

bottleneck, ambivalent but pragmatic attitudes to AI, and constraints of current institutional guidance). Themes were developed iteratively by re-reading the transcripts, reviewing and clustering codes, and discussing candidate themes between the first and second author, with attention to how well they captured SSWS perspectives in a coherent and practically useful way. Table 1 summarises the final themes, brief defi-

Table 1 Summary of themes from SSWS interviews about students with dyslexia

Theme	Sub-themes	Brief description	Representative quotation
Academic writing and expression	Expressing ideas in academic English; structuring assignments; using appropriate tone and vocabulary	Students often have sound ideas but struggle to express them in formal academic writing, so their work can appear simplistic or disjointed relative to their understanding.	“They know the material when they talk, but their written work sometimes does not do them justice.”
Critical reading and argumentation	Evaluating sources; adopting a critical stance; constructing arguments	The cognitive load of reading dense material can leave little capacity for critical evaluation, so students may accept sources at face value or hesitate to challenge them.	“By the time they have decoded the text, there is not much energy left for critiquing it.”
Planning, organisation and time management	Breaking tasks down; estimating time; avoiding last-minute rush	Many students struggle to plan ahead, break assignments into smaller steps and estimate how long stages will take, leading to rushed work and inefficient patterns of study.	“They often leave things until the last minute because it is hard to see all the steps involved.”
Use of assistive tools and adoption barriers	Awareness and training; effort to learn tools; experimentation versus routine	Uptake of institutional tools is patchy: some students are unaware of features, others abandon tools when benefits are not immediate, while a few continually experiment without forming a consistent strategy.	“We see students who try every new app under the sun, but they are spinning their wheels rather than building a consistent strategy.”

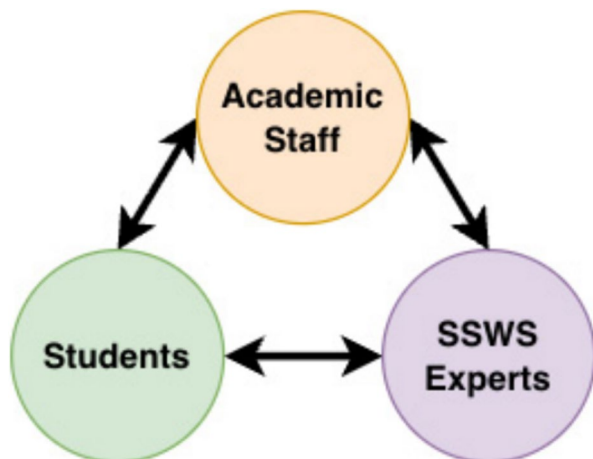
nitions and illustrative quotations. We did not use numerical stopping rules or claim data saturation in a quantitative sense; given the small, expert sample, the aim was to generate a rich, situated account to inform framework design rather than an exhaustive mapping of all possible viewpoints.

For the quantitative strand, closed-response survey items were summarised using descriptive statistics (frequencies and percentages). Percentages were calculated using the number of valid responses for each item as the denominator, excluding missing responses. For multiple-response items, percentages represent the proportion of respondents selecting each option, so totals can exceed 100%. We did not apply minimum frequency thresholds or percentage cut-offs in reporting closed-response items. Open-ended responses were analysed qualitatively. They were coded in a light-touch way, using categories aligned with the main interview themes to support comparison between staff and student perspectives, and to provide illustrative context alongside the descriptive survey patterns. No inferential statistical analyses were conducted due to the small, self-selected sample; survey findings are used to complement and contextualise the qualitative themes rather than as a basis for strong generalisations. Because the number of respondents drawing on disability-related support was small and diagnostic information incomplete, we did not conduct formal statistical comparisons between dyslexic and non-dyslexic students; instead, the survey is used to provide descriptive context for the qualitative findings and to inform the design of the framework.

3.4 Development of the AI integration pedagogical system

Following data collection and analysis, we moved into the design phase, focused on creating a structured AI integration pedagogical system to guide responsible AI use in academic work. The workflow was informed by challenges and needs identified in both the literature review and our empirical findings. It explicitly considered the three key stakeholders and their interactions (Fig. 2): academic staff, who specify permitted AI use; students, who follow the workflow for those use cases; and SSWS, who

Fig. 2 Key stakeholders and validation stages in the framework development process



support students in applying the workflow and clarify any ambiguities with academic staff.

We adopted a principle of alignment with existing academic processes, structuring the workflow around the typical stages of completing an assignment (understanding the brief and readings, generating ideas, conducting research, outlining, drafting and proofreading). For each stage we considered: (1) which AI-powered tools or features could provide support, (2) what steps students should take to use the tool effectively and (3) what safeguards or reflective prompts were needed to ensure ethical use. The outcome was a draft pedagogical system comprising a staff-facing component (the AI Engagement Spectrum) and a student-facing component (the AI Scaffold), which were then refined through iterative feedback from SSWS and review against UDL and constructive alignment principles.

3.5 Validity and trustworthiness

Validity and trustworthiness were considered throughout the study. Content validity for the interview guide and survey was supported by grounding instruments in the literature on dyslexia, UDL and AI-supported study, and by consultation with SSWS. Thematic analysis incorporated peer debriefing within the research team, and we maintained an audit trail of coding decisions and theme revisions to support dependability and confirmability (Shenton, 2004). Survey data are treated as exploratory; the small, self-selected sample and lack of inferential statistics mean that findings are interpreted cautiously, primarily as context for the qualitative insights and as inputs into the design of the framework. We therefore do not claim theoretical or thematic saturation; instead, the qualitative strand is positioned as a focused, context-specific account that supports the design and refinement of the framework, with findings treated as exploratory and transferable rather than statistically generalisable.

3.6 Ethical considerations

All procedures complied with institutional ethical guidelines and received approval from the relevant university ethics committee. Participants received an information sheet and provided informed consent prior to taking part. Interview data were anonymised during transcription, and potentially identifying details were removed or generalised. Survey participation was voluntary and anonymous, with no requirement to disclose disability status. Items that could be considered sensitive, such as questions about the use of support services, are reported only in aggregate to minimise any risk of indirect identification in a small sample. Participants were informed that they could withdraw from the study at any time before data aggregation without penalty.

4 Findings from interviews and student survey

This section reports the key qualitative and quantitative findings from the SSWS interviews and student survey that informed the design of the *AI Scaffold* framework and *AI Engagement Spectrum*. The results are organised thematically: Section 4.1 focuses on the academic challenges faced by students with dyslexia in academic tasks (RQ1); Section 4.2 examines current use of and adoption barriers around assistive tools and AI-powered technologies (RQ1); and Section 4.3 explores students' and staff's attitudes towards GenAI, including perceived benefits and concerns (RQ2). A final subsection (Section 4.4) integrates these strands to show how the combined evidence shaped the framework (RQ3).

4.1 Challenges faced by students with dyslexia in academic tasks

The interviews with SSWS reaffirmed many challenges documented in the dyslexia literature while adding detail about how they play out in day-to-day university work. Advisers described students who often have strong ideas and oral contributions but struggle to convey these effectively in writing, especially in longer, formally assessed assignments. Reading demands were reported as a constant pressure point: students frequently fall behind on text-heavy modules and find it difficult to identify what is essential in dense readings, which in turn affects their confidence in seminars and exams. These challenges are compounded by organisational difficulties, such as breaking large assignments into manageable steps and sustaining attention across multiple online platforms. Table 1 summarises the main themes that emerged in relation to academic reading and writing, critical thinking, planning and organisation, and the use of assistive tools.

Taken together, these themes suggest that many students with dyslexia arrive at SSWS with well-developed ideas and insights but lack confidence in managing the reading and writing demands of their programmes. Advisers noted that students often avoid critical-analytical tasks or leave them to the last minute because they find it hard to translate broad insights into structured arguments under time pressure. Planning and organisation difficulties mean that assignments can remain at the level of notes or partial drafts for too long, contributing to elevated stress and lower attainment. Staff also highlighted that students' use of assistive tools is frequently reactive and fragmented rather than embedded in coherent study routines. These observations directly informed the design priorities for the *AI Scaffold*, which aims to support reading, planning, drafting and checking as an integrated workflow rather than as isolated one-off actions.

4.2 Use of assistive tools and adoption barriers

Staff interviews and student survey responses indicated that many students with dyslexia have access to a range of assistive tools, including TTS, dictation, screen readers, grammar checkers, mind-mapping software and note-taking apps. However, use was often described as patchy and inconsistent rather than embedded as part of everyday study. Some students reported relying heavily on a single tool (for example, TTS

for reading or Grammarly for surface-level checking), while others experimented briefly with several tools and then abandoned them. Only a minority appeared to have developed stable routines in which assistive technologies were integrated across modules and assessment types.

Both staff and students identified barriers that help explain this pattern. Practical issues included fragmented interfaces, login friction and the need to switch between multiple platforms while working in the virtual learning environment. Some tools were only available on campus machines or through limited licences, constraining their usefulness for home study. Training was another recurrent theme: students often received a one-off demonstration but little follow-up support, leaving them unsure how to adapt tools to different tasks. Stigma also featured strongly in the interviews; several advisers noted that students were reluctant to be seen using “special” software in public spaces or to disclose their use of assistive technologies to peers.

The student survey echoed these concerns. Respondents who reported low or sporadic use of assistive tools commonly cited not knowing which tools would genuinely help, finding interfaces confusing, or feeling that experimentation took too much time during busy semesters. Some perceived a mismatch between tools and assessment demands, describing situations in which technologies helped with reading or drafting but did not seem obviously applicable to exams or group projects. Collectively, these findings suggest that the problem is not a lack of tools but a lack of structured, context-sensitive guidance on how to weave them into coherent study habits and assessment workflows.

4.3 Attitudes towards AI tools: Benefits and concerns

A significant part of our investigation focused on how students and staff currently perceive AI tools in higher education (RQ2). Survey responses indicated that most students in our sample had experimented with AI in some capacity (91.9%), with only a small minority reporting no use. This high level of exposure echoes national findings, such as the Higher Education Policy Institute (HEPI) report by Freeman (2025), which reports that more than half of UK undergraduates have used generative AI tools to help with assessments. In our data, AI was most commonly used for summarising readings, brainstorming ideas, overcoming writer’s block and checking grammar and phrasing, with fewer students using AI to structure assignments (Fig. 3).

Qualitative comments and SSWS interviews added nuance. Students described AI as a convenient, non-judgemental resource that could help them “get started”, clarify unclear concepts and produce first drafts for later editing, while staff highlighted benefits for reducing blank-page anxiety and providing alternative explanations. At the same time, both groups stressed that these benefits depended on students maintaining a critical stance and treating AI outputs as provisional rather than authoritative. Concerns centred on over-reliance, accuracy and academic integrity. Many students worried that heavy use of AI could weaken their own skills or make them “too lazy to think”, and some were anxious that AI-generated content might be considered plagiarism. Uncertainty about institutional expectations was common: a substantial

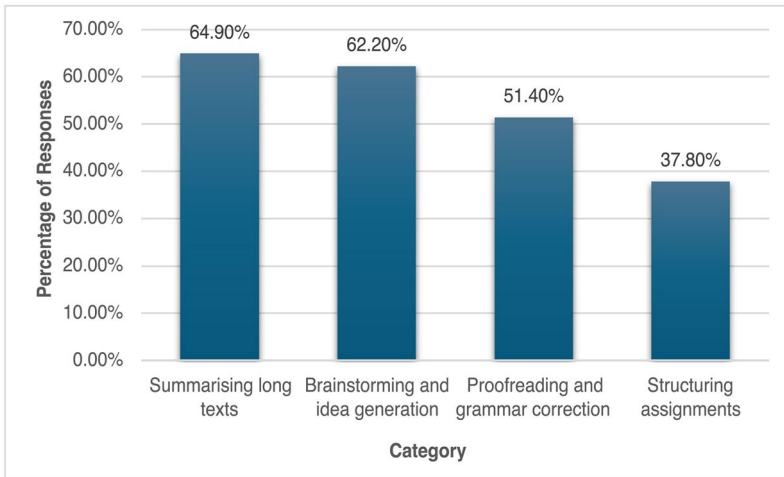


Fig. 3 Purposes for which students use AI tools

proportion of respondents were unsure what level of AI assistance was acceptable. Figure 4 shows the most frequently selected concerns.

Staff interviews reflected this ambivalence. Some tutors were hesitant to mention AI at all, fearing that any suggestion could encourage misconduct, while others saw clear potential but were anxious about detection, fairness and workload. Overall, the findings indicate that students and staff are navigating AI in a climate of cautious experimentation and uncertainty, reinforcing the need for task-level guidance and transparent, documented workflows, as developed in Section 5.

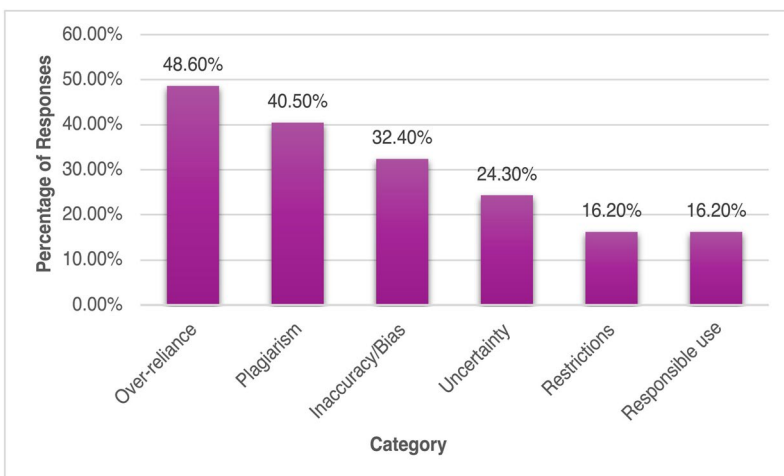


Fig. 4 Students' primary concerns about using AI tools in academic work

4.4 Integration of qualitative and quantitative findings

The qualitative and quantitative strands converged on a broadly consistent picture. Staff interviews offered a detailed, practice-based view of the challenges faced by students with dyslexia, their uneven use of assistive tools and the opportunities and risks associated with AI, while the student survey indicated high levels of AI exposure, frequent use of grammar and spelling tools and mixed uptake of more specialised assistive technologies. Students' reports of relying on a small subset of tools, experimenting briefly with others and feeling unsure how to use AI responsibly closely echoed the concerns raised by SSWS advisers. Table 2 summarises these points of convergence and divergence and the resulting implications for the framework.

Table 2 Integrated summary of staff and student perspectives and implications for the framework

Theme / issue	Qualitative insight (SSWS interviews)	Quantitative pattern (student survey)	Implications for <i>AI Scaffold</i> and <i>AI Engagement Spectrum</i>
Academic writing and expression	Staff emphasised that students with dyslexia often have strong ideas but struggle to express them in formal academic writing, which can under-represent their understanding.	Most respondents reported using AI to summarise texts, brainstorm ideas and improve writing clarity and grammar, indicating that they already seek support around heavy reading and written expression.	The Scaffold emphasises AI support at the planning and drafting stages (for example summarising, idea generation and language refinement) while preserving student ownership through explicit authorship checks and reflection prompts.
Planning, organisation and time management	Interviews highlighted difficulties in breaking tasks into manageable steps and estimating how long work will take, contributing to last-minute rushing and inefficient study patterns.	Several students described irregular use of planners and study-support tools, with comments such as having software installed but “forgetting to use it”, illustrating habit-formation challenges.	The Scaffold provides a step-by-step workflow that breaks assignments into smaller actions and prompts students to plan when and how they will use both assistive and AI tools, making planning itself a visible study behaviour.
Use of assistive and AI tools	Staff observed patchy adoption of institutional tools and, at the other extreme, students experimenting with many apps without developing a consistent strategy.	Survey responses showed high experimentation with GenAI tools but varied depth of use, with some students mainly “trying things out” rather than integrating tools into regular study routines.	The Scaffold helps students move from ad hoc experimentation to deliberate, goal-oriented tool use by linking each AI use case to a specific study task and intended learning outcome.
Integrity, over-reliance and policy clarity	Both staff and students expressed concern that AI could erode skills or cross into misconduct if used uncritically or without clear rules, and tutors stressed the need for students to remain “critical beings”.	Around half of respondents reported worries about over-reliance and plagiarism, and a substantial minority were unsure what level of AI assistance is permitted under university rules.	The Engagement Spectrum supports staff in defining assignment-specific boundaries, while the Scaffold requires students to document how they have used AI, making expectations and accountability explicit and reducing uncertainty about acceptable practice.

Each strand also contributed something the other could not provide alone. Staff perspectives foregrounded issues of workload, stigma and policy, showing how inconsistent guidance and assessment design can leave students uncertain about expectations. Survey responses illustrated how these uncertainties play out at scale: many students were enthusiastic about AI's potential to reduce reading and writing barriers but anxious about "crossing a line" that might be considered misconduct. This integrated picture directly informed the dual-component system developed in the next section. The *AI Engagement Spectrum* addresses the need for clear, task-level decisions about permissible AI use, while the *AI Scaffold* responds to documented difficulties with reading, planning, drafting and checking by organising AI and non-AI supports into a structured workflow that students can apply across modules.

5 A novel, dual-component pedagogical system for responsible AI integration

Grounded in the literature review and the mixed-methods findings, this section introduces a dual-component pedagogical system for responsible AI-supported assessment. The instructor-facing *AI Engagement Spectrum* applies constructive alignment to set task-specific rules for permissible AI use, while the student-facing *AI Scaffold* offers a UDL-aligned workflow that guides students with dyslexia through reading, planning, drafting and revising with AI support. Together, these components respond to RQ3 by linking institutional policy, pedagogical intent and day-to-day academic work, so that decisions about AI use become visible as study activities and process artefacts within a single AI-aware assessment system rather than an add-on policy layer.

5.1 The *AI Engagement Spectrum* and establishing permissible use

Before students apply the *AI Scaffold* to an assignment, the lecturer must define clear rules of engagement for that task. This professional judgement is guided by the Intended Learning Outcomes (ILOs): which skills must students demonstrate unaided, and where can AI legitimately scaffold learning without undermining those goals? The *AI Engagement Spectrum* supports this decision by grouping AI assistance into five use cases that can be stated directly in the assignment brief.

Within constructive alignment, the Engagement Spectrum and the Scaffold link outcomes, activities and assessment. ILOs specify the cognitive behaviours a task should elicit (for example explain, analyse or create); the Engagement Spectrum turns these verbs into rules about which AI functions may support but not replace those behaviours, and the *AI Scaffold* organises students' study activities so that they practise the intended skills while generating process evidence alongside the final product. This makes the rationale for permitting or constraining particular AI uses explicit to staff and students.

5.1.1 The theoretical foundation of the five AI engagement use cases

The five use cases of the *AI Engagement Spectrum* (Idea Generation, Research, Structuring, Drafting and Proofreading) draw on process models of writing. Flower and Hayes (1981) conceptualise writing as a non-linear cycle of planning, translating and reviewing, while process-writing approaches distinguish pre-writing, drafting, revising and editing. Idea Generation, Research and Structuring align with planning stages; Drafting with translating ideas into text; and Proofreading with reviewing and editing. This non-linear view justifies the flexible, cyclical structure of the *AI Scaffold*, which allows students to move between stages rather than follow a rigid checklist.

5.1.2 Academic perspective on the five AI engagement use cases

The lecturer must decide which AI functions would scaffold learning versus which would perform the core cognitive task for the student, risking the assessment's validity (Stokkink, 2025). To help lecturers make these informed decisions, the spectrum is organised around five common use cases identified in Section 5.1.1, each representing a key stage in the academic writing process. The following subsections detail each of these. For each one, we will first define its specific **Function**, describing how AI can be used at that stage, and then outline the **Lecturer's Consideration**, which provides the pedagogical rationale for either permitting or restricting its use based on the assignment's intended learning outcomes.

Use Case 1: AI for idea generation and brainstorming Function: Using AI as a creative partner to brainstorm topics, overcome writer's block, formulate initial questions, or explore potential arguments.

Lecturer's Consideration: This is often permitted as it supports the initial, divergent thinking phase (a key part of Bloom's create level) of writing without substituting for the core analytical work to follow, unless independent idea innovation skills are a core component in the assessment.

Use Case 2: AI for research Function: Using AI to identify relevant search terms, find potential academic sources, or generate summaries of dense articles to aid comprehension.

Lecturer's Consideration: Permissible when the ILO focuses on synthesis or application (Bloom's apply or analyse), but may be restricted if the ILO is to develop the skill of literature searching itself (which relates to Bloom's remember and understand).

Use Case 3: AI for structuring and outlining Function: Using AI to organise brainstormed ideas or research notes into a coherent outline or structure for an essay or report.

Lecturer’s Consideration: A powerful scaffold for students who struggle with linear planning. This is often allowed, as the student must still populate the structure with their own analysis and prose (Bloom’s analyse and apply).

Use case 4: AI for drafting Function: Using AI to expand the student’s own outline points into initial draft paragraphs or to rephrase sentences for clarity and academic tone.

Lecturer’s Consideration: This is the most contentious area. It should only be permitted when the ILO is not focused on the student’s ability to construct prose or formulate arguments from scratch (a core component of Bloom’s higher-order skills, like analyse, evaluate, and create). If allowed, it must be paired with a strict requirement for the student to heavily revise, personalise, and take full authorial ownership of the text.

Use Case 5: AI for Proofreading and Revision Function: Using AI to check for grammatical errors, spelling mistakes, and issues with sentence clarity in a completed draft.

Lecturer’s Consideration: This is commonly permitted, as it helps polish the final work without compromising the student’s intellectual contribution which occurs at the higher levels of Bloom’s Taxonomy. However, local policy may vary and transparency requirements apply (Cornell University Center for Teaching Innovation, 2023; Vanderbilt University Center for Teaching, 2023).

Operationally, permissions should track the cognitive level specified in the ILOs. If the intended verbs emphasise lower-to-mid levels (e.g., identify, explain, apply), brainstorming or outlining support may be allowed while drafting is constrained; where ILOs target higher-order outcomes (e.g., analyse, evaluate, create), students must demonstrate those behaviours themselves and any AI use must be bounded by explicit authorship checkpoints (Anderson & Krathwohl, 2001; Biggs et al., 2023).

As a practical illustration, a module handbook might include language such as: ‘In this assignment you may use GenAI tools to brainstorm ideas (Use Case 1), explore search terms and summarise sources (Use Case 2), and proofread your final draft (Use Case 5). You must not use AI to generate full paragraphs or restructure the assignment for you, and you remain responsible for the accuracy and originality of all submitted work. Any AI use must be logged in your process notes as described in the assignment brief.

By explicitly stating which of these five use cases are permitted for an assignment, the lecturer provides the “safe passage” that students need to engage with AI confidently and ethically. This act of defining permissible use transforms the *AI Scaffold* from a generic guide into a context-specific tool, ensuring that technology serves, rather than undermines, the fundamental goals of learning. For instance, a lecturer might state in a module guide: “For this assignment you may use GenAI tools for idea generation and proofreading only. You must not use AI to produce an outline or draft paragraphs, as these stages are part of the learning outcomes being assessed. Any AI use should be recorded in a short appendix that lists the tools used, the prompts you

entered, and how you checked the outputs for accuracy.” In practice, staff can adapt this template by first selecting the permitted Engagement Spectrum use cases and then adding one or two simple checkpoints they expect students to evidence.

5.2 The AI Scaffold framework and detailed walkthrough

Once an instructor has established the pedagogical prerequisites using the *AI Engagement Spectrum*, the *AI Scaffold* provides students with a structured workflow for engaging with AI responsibly. Students with dyslexia can use it with the help and support of SSWS. This entire process is visually represented in the flowchart in Fig. 5 which provides a high-level map of the system. The reader is encouraged to see this figure as an overview, as a detailed, magnified view of each stage will be presented after the walkthrough. The framework is not a simple technical guide, but a pedagogical approach that applies UDL principles (offering multiple means of representation, action, and engagement), emphasises process over product, and embeds human-in-the-loop agency at every stage.

Although the scaffold can support general study skills, it is primarily intended to be integrated into assessed coursework and project work. It scaffolds five interconnected aspects of the academic workflow, reading and summarising, note-making, planning and structuring, drafting, and proofreading/revision, each paired with

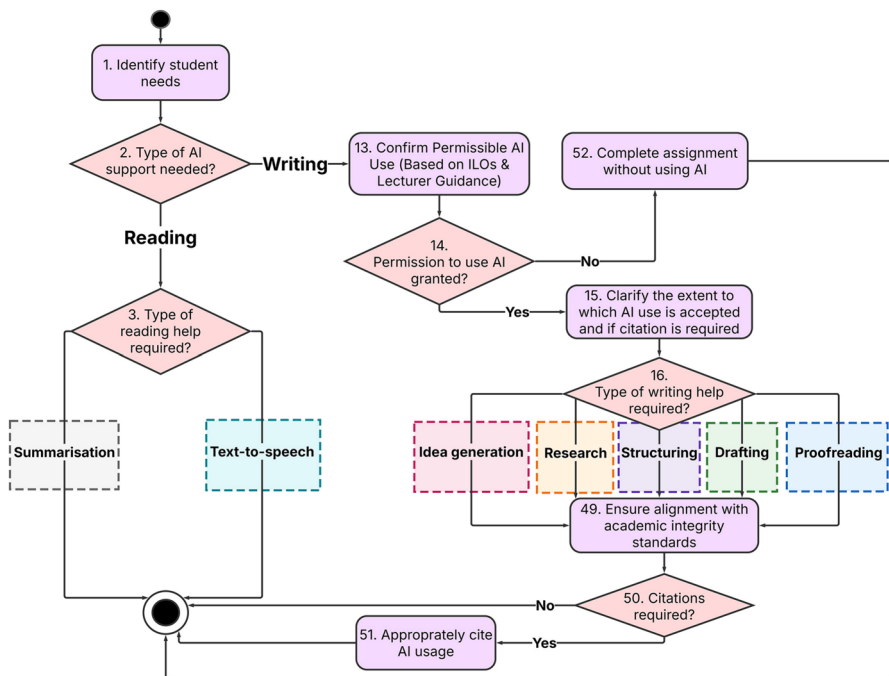


Fig. 5 The AI integration workflow. This flowchart illustrates the major stages (Reading and writing) and how AI can be used in each, under certain conditions. Key decision points (diamond shapes) prompt the student to check permissions (e.g. “Is AI use allowed?”). The detailed AI Engagement Use Cases can be found in Fig. 6

explicit checkpoints that require students to interpret assignment briefs, make judgments, and generate their own text. AI tools are positioned as cognitive supports that reduce extraneous load (for example decoding dense texts or spotting surface-level errors) while leaving the core analytical and creative work with the student, directly addressing the concerns about over-reliance, academic integrity and policy uncertainty reported earlier in the paper.

As shown in Fig. 5, the framework is intentionally tool-agnostic and structured around two core pathways that mirror the academic cycle: the Reading Support Pathway (RSP) and the Writing Support Pathway (WSP). It positions the student as the final decision-maker of meaning and authorship, embedding critical checkpoints to ensure human-in-the-loop agency. The following sections will deconstruct this process, providing a detailed walkthrough of each pathway.

5.3 The RSP, scaffolding comprehension and critical engagement

This pathway is designed to help students manage and understand source material, directly addressing the significant challenges faced by learners with dyslexia in text-heavy academic environments.

5.3.1 Providing multiple means of representation

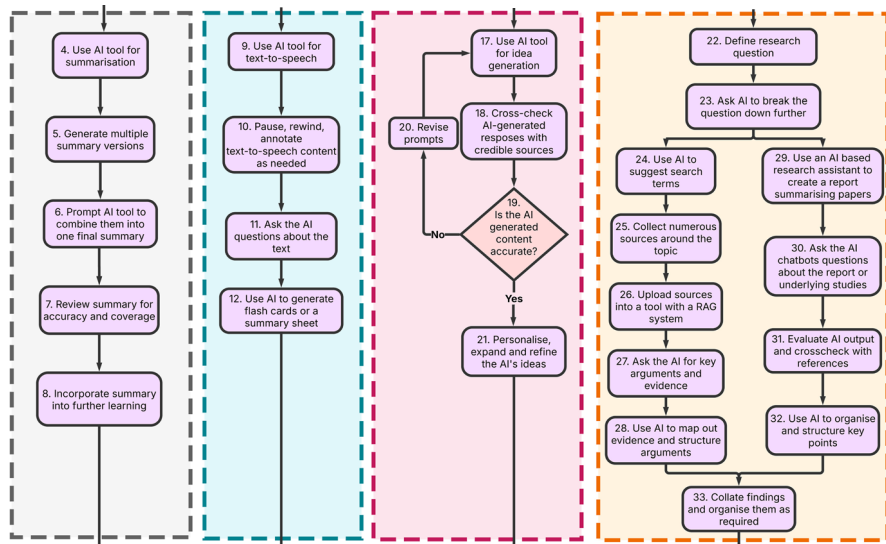
Academic success often depends on processing large volumes of dense text, which can be particularly challenging for students with dyslexia because decoding is slow, fatiguing and cognitively demanding. The RSP offers two main interventions that provide multiple means of representation, a core UDL principle.

First, the framework supports the use of AI-powered summarisation. This intervention directly addresses the challenge of cognitive load by condensing lengthy articles or book chapters into digestible key points. This helps students focus their limited working memory resources on comprehension and analysis rather than on initial processing. The high demand for this function is confirmed by the survey data indicating that summarising texts is the single most common use of AI among students, reported by 65% of respondents.

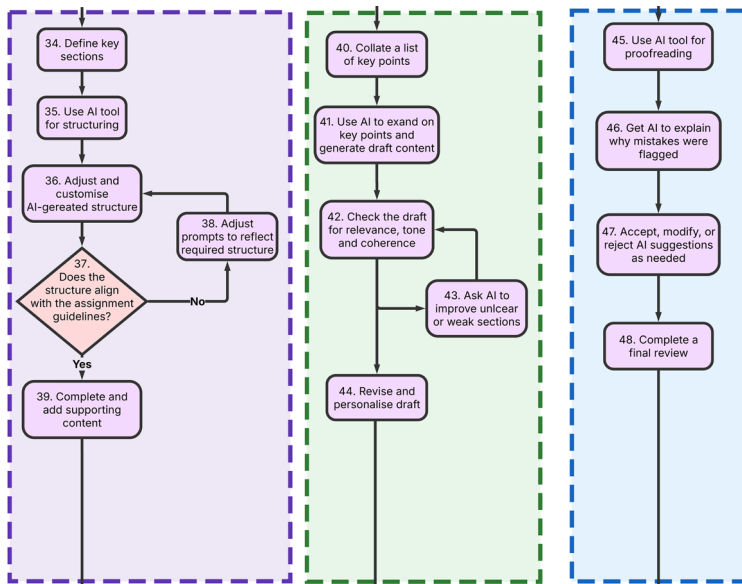
Second, it recommends the use of TTS technology, which uses auditory processing to bypass the phonological decoding barrier that is a hallmark of dyslexia. This allows students to access information through listening, which can be a more efficient and less cognitively taxing channel. Together with summarisation, this supports both perception and building knowledge by highlighting patterns and critical features. The recommended workflows for both TTS and AI-powered summarisation are visualised in Fig. 6a and b, respectively. When summarisation is permitted under the *AI Engagement Spectrum*, students should treat it as a comprehension aid, not a replacement for reading, and proceed directly to the accuracy, and coverage checkpoint.

5.3.2 Ensuring active and critical analysis

While TTS and summarisation tools can substantially reduce decoding load, they also risk encouraging uncritical cognitive offloading, where students outsource eval-



(a) Summarisation - RSP (b) TTS - RSP (c) Idea Generation - WSP (d) Research - WSP



(e) Structuring - WSP (f) Drafting - WSP (g) Proofreading - WSP

Fig. 6 Closer overview of the *AI Engagement Spectrum* use cases shown in Fig. 5

uation to the tool and accept AI summaries without engaging deeply with the source material (Chiang, 2024; Risko & Gilbert, 2016). To counter this, the Reading Support Pathway includes a non-negotiable checkpoint to review summary for accuracy and coverage. At this stage students (1) compare the AI summary with the original text, (2) check for omissions, distortions or bias, and (3) relate the summary back to their task and prior knowledge. This simple routine keeps the student in an active, evaluative role rather than as a passive consumer of AI output.

5.4 The WSP, a staged approach from conception to completion

The WSP breaks the often overwhelming process of academic writing into manageable stages with optional AI support. The first and most critical step is for the student to consult and understand the rules of engagement for AI on that particular assignment, as defined by the lecturer using the *AI Engagement Spectrum*. This initial checkpoint (Step 15 in Fig. 5) acts as the gatekeeper for the whole process: a student will only engage with those use cases that the instructor has permitted. Once permissions are clear, the student can follow the relevant stages below and move back and forth between them as needed.

5.4.1 Use case 1: Idea generation and brainstorming

Early writing stages are often the hardest, particularly for students who struggle to turn holistic ideas into a concrete starting point. Here AI acts as a creative partner, helping students brainstorm topics, angles and examples and thereby lowering the barrier of writer's block. In our survey, 60% of respondents reported using AI for this purpose. This stage is governed by the checkpoint *Is the AI-generated content accurate?*, which prompts students to discard factually incorrect or irrelevant suggestions before investing further time in them. The corresponding workflow is illustrated in Fig. 6c.

5.4.2 Use cases 2 & 3: Research and structuring

Once an idea is chosen, students move between research and structuring. AI support at this stage can include breaking a broad question into sub-questions, suggesting search terms, extracting key points from sources and proposing a tentative outline. We treat research and structuring together because externalising an outline is part of effective planning and is associated with improved writing quality (Graham & Perin, 2007). In practice, students iterate between gathering evidence and organising their argument, so the framework encourages them to refine AI suggestions, check claims against original sources and adapt the outline to their own stance and the assessment brief, as shown in Fig. 6d and e.

5.4.3 Use case 4: Drafting with AI and asserting authorship

This stage addresses the most contentious use of AI in academia: text generation. Here the framework positions AI as a writing assistant that expands the student's

own outline points into provisional paragraphs, rather than as a co-author. This can be especially helpful for students who struggle with formal academic expression or for students with dyslexia who find writing slow and effortful. Crucially, this support is immediately followed by a mandatory step to revise and personalise the draft. At this point students must rewrite the text in their own academic voice, integrate their own insights and course knowledge, and ensure alignment with the assignment brief and overarching argument. This is the moment where intellectual and authorial ownership is actively asserted.

The step ensures that higher-order processes (analyse, evaluate, create) are enacted by the student, not the tool, which is essential when ILOs target those levels (Anderson & Krathwohl, 2001; Biggs, 1996). It is the primary defence against both misconduct and skill atrophy, ensuring that although AI assisted initial sentence construction, the final work is substantively the student's own. Figure 6f provides a step-by-step guide to this drafting and revision process, emphasising the critical loop of student authorship.

5.4.4 Use case 5: Elevating proofreading to a learning opportunity

The final use case employs AI for the widely accepted task of proofreading for grammar, spelling and style. The framework introduces a key pedagogical step that turns this from a simple corrective activity into an opportunity for deeper learning: students ask the system to explain why each change is proposed and then accept, modify or reject it. This guided metacognition requires students to articulate why a suggestion is correct before accepting it, supporting durable learning through desirable difficulties (Bjork, 1994; de Bruin, 2023). It interrupts the passive cycle of simply accepting corrections and instead prompts engagement with the underlying rule or principle, turning proofreading into a brief episode of active learning and self-reflection. This metacognitive approach is outlined in the flowchart in Fig. 6g.

5.5 A synthesis of the *AI Scaffold*, a practical guide for students, academics and SSWs

The preceding sections have deconstructed the *AI Scaffold*, detailing each stage of reading and writing with corresponding visual guides. Table 3 synthesises this information into a single, accessible resource by mapping each stage to dyslexia-related challenges, recommended AI-supported actions, human-in-the-loop checkpoints and primary UDL principles. For students with dyslexia, this explicit pairing of tools and checkpoints is intended to reduce working-memory load, support planning and self-monitoring, and build confidence in producing extended academic text.

In practice, brief training and ongoing reminders are needed to sustain these routines; without support, assistive tools can be under-used or discontinued over time (Bäck et al., 2024; Pino & Mortari, 2014). We therefore recommend introducing the scaffold in class, sharing Table 3 as a handout, and revisiting the checkpoints during supervision or study-skills appointments.

This guide serves as a shared resource for students, academic advisors and faculty. By explicitly linking dyslexia-related challenges to recommended AI-supported

Table 3 A practical synthesis of the *AI Scaffold* framework

Framework Stage	Dyslexia-Related Challenge Addressed	Recommended AI-Supported Action	Critical Checkpoint / Human-in-the-Loop Action	Primary UDL Principle Alignment
Reading & Summarising	Slow decoding; reading fatigue; cognitive load from dense text.	Use TTS for auditory input. Use AI to generate a summary of key points.	Review AI summary against the original text for accuracy and coverage. Ask clarifying follow-up questions.	Representation (Providing multiple ways to access information)
Idea Generation	“Writer’s block”; difficulty initiating tasks; holistic thinking needs a starting point.	Use AI as a creative partner to brainstorm topics, arguments, or examples.	Cross-check AI-generated ideas for factual accuracy and relevance. Select and adapt ideas, rather than accepting them wholesale.	Engagement (Optimising individual choice and autonomy)
Research & Structuring	Difficulty breaking down tasks; translating holistic ideas into a linear plan.	Use AI to generate search terms, extract key findings from sources, and create a potential outline.	Critically evaluate all AI-generated points against original sources. Revise and refine the AI-suggested structure to align with the assignment brief and personal argument.	Action & Expression (Supporting planning and strategy development)
Drafting	Difficulty with formal academic expression; slow writing/typing speed.	Use AI to expand on the student’s own outline points to create initial draft paragraphs.	THE CRITICAL STEP: Revise, rewrite, and personalise the entire draft in the student’s own voice. Add unique insights and examples.	Action & Expression (Facilitating expression and communication)
Proofreading & Final Checks	Surface-level errors in grammar and spelling; lack of confidence in final submission.	Use AI for grammar, spelling, and style checks. Prompt the AI to explain why an error was flagged.	Actively accept, modify, or reject AI suggestions. Perform a final, independent read-through. Ensure final submission aligns with academic integrity policies and AI use is declared if required.	Engagement (Supporting self-assessment and reflection)

actions, human-in-the-loop checkpoints and UDL principles, it makes the framework’s design transparent and actionable. Rather than offering isolated tips, it presents a coherent system in which AI support, human judgement and accessibility commitments are tightly connected. To our knowledge, this is the first approach that couples an ILO-aligned permission model with a UDL-aligned student workflow for AI use in higher education.

6 Discussion

This section interprets the findings in light of the wider literature and the theoretical framework developed in this study. The empirical work addressed our initial research questions by identifying specific challenges for students with dyslexia (RQ1) and mixed feelings about AI (RQ2), which in turn informed the design of the two-part framework (RQ3). The following subsections set out the main theoretical contri-

butions, beginning with the overarching conceptual model that emerged from the research.

6.1 The nested scaffolding model as a transferable architecture for responsible AI integration

The framework is best understood as part of a multi-level system of Nested Scaffolding rather than as a single tool. This model offers a transferable architecture for responsible AI integration with four interconnected layers: institutional, pedagogical, procedural and cognitive.

1. **Institutional scaffold:** University-level policies on academic integrity, data privacy and ethical technology use set broad principles such as transparency and attribution (Cornell University Center for Teaching Innovation, 2023).
2. **Pedagogical scaffold:** Lecturers interpret these principles in light of constructive alignment, analysing ILOs and defining the permissible scope of AI use for each assessment (Biggs, 1996; Stokkink, 2025).
3. **Procedural scaffold:** The AI Scaffold operates at this level, giving students a step-by-step workflow for responsible AI use within those task-specific boundaries.
4. **Cognitive scaffold:** AI tools provide targeted support for sub-tasks such as summarisation or brainstorming, reducing extraneous load while leaving core reasoning and authorship with the learner (CAST, 2024).

Taken together, these layers show how institutions, academics and students can coordinate AI integration so that it remains ethical, transparent and aligned with core learning objectives.

6.2 Synthesising flexibility and constraint in AI pedagogy with UDL and constructive alignment

A key theoretical contribution is the synthesis of UDL and constructive alignment in AI-assisted pedagogy. UDL provides the imperative to offer flexible routes into learning and multiple ways of engaging with and expressing understanding (CAST, 2024), while critics warn that it can be interpreted as a licence to remove all challenge or desirable difficulty (Bjork, 1994; Boysen, 2024; de Bruin, 2023). GenAI intensifies this tension because it can both lower access barriers and displace core cognitive work.

Constructive alignment helps to resolve this by tying AI permissions to the cognitive verbs in the ILOs (for example analyse, evaluate or create) (Anderson & Krathwohl, 2001; Biggs, 1996). This logic underpins the AI Engagement Spectrum: it supports flexibility where AI can legitimately scaffold access and expression, and constraint where AI would replace the skills an assessment is intended to measure. Table 4 offers a practical heuristic for applying this principle, while the UDL mapping in Table 5 in Appendix C shows that the framework as a whole remains systematically aligned with UDL 3.0 rather than being an ad hoc collection of tips.

Table 4 A Constructively aligned model for permissible AI Use in academic writing

ILO & Bloom's Verb	Example student task	Desirable difficulty (Core Cognitive Skill to be Assessed)	Recommended AI permission level	Pedagogical rationale (Linking AI Use to ILO)
Remember/ Understand: Identify, list, summarise, explain.	Summarise the key arguments of a provided research paper.	Comprehension and extraction of core concepts.	Assistive Use Permitted: AI can be used for summarisation, rephrasing complex sentences, and generating key points.	AI removes the extraneous load of decoding dense text, allowing the student to focus cognitive resources on the desirable difficulty of comprehending and synthesising the main ideas (Jose et al., 2025).
Apply: Implement, use, solve.	Use a specific theoretical framework from class to analyse a new case study.	Application of a known procedure or concept to an unfamiliar context.	Scaffolded Use Permitted: AI can be used for brainstorming initial connections or structuring the analysis, but not for generating the analysis itself.	AI can help overcome initial organisational hurdles, but the core cognitive work of applying the framework, the desirable difficulty, must be performed by the student to demonstrate the ILO (Stokkink, 2025).
Analyse: Differentiate, organise, compare, contrast.	Compare and contrast the methodologies of two conflicting studies.	Deconstruction of information into constituent parts to explore relationships.	Limited Use Permitted: AI can be used for initial research or to generate a list of potential comparison points. The analysis and argument must be the student's own.	The desirable difficulty is the critical analysis and identification of nuanced differences. Overuse of AI would offload this core skill. The student must demonstrate their own analytical process.
Evaluate: Justify, critique, defend, appraise.	Write a critique of a published author's central argument.	Making judgments based on criteria and standards through checking and critiquing.	Highly Restricted/ Brainstorming Only: AI can be used to generate potential counterarguments for the student to consider and rebut, but not to write the critique.	The ILO is explicitly about the student's ability to make a judgment. This is a core higher-order thinking skill that must be demonstrated independently. AI is used only as a sparring partner (Stokkink, 2025).
Create: Generate, design, produce, formulate.	Formulate a novel research question and propose an original research design.	Putting elements together to form a coherent or functional whole; reorganising elements into a new pattern or structure.	Prohibited (for core task): AI use is prohibited for generating the research question and design. It may be permitted for subsequent proofreading of the final proposal.	The essence of this ILO is originality and synthesis. Allowing AI to generate the core creative output would fundamentally undermine the assessment's validity and prevent the development of this key skill.

6.3 Moving from assistive technology to assistive intelligence by preserving desirable difficulties

The framework reframes GenAI tools as forms of “assistive intelligence” rather than simple assistive technologies. Traditional tools such as TTS mainly remove surface-level barriers in accessing or producing text, whereas GenAI can generate ideas, outlines and prose. Without guidance, this risks bypassing the cognitive work that

assessments are intended to measure and can encourage over-reliance on automation (Ji et al., 2023; Kasneci et al., 2023; Risko & Gilbert, 2016).

The AI Engagement Spectrum and AI Scaffold take a different stance. By requiring lecturers to decide which stages must remain unaided, and by embedding checkpoints that ask students to verify AI outputs, justify changes and rephrase material in their own words, the framework preserves productive “desirable difficulties” that support durable learning (Bjork, 1994; de Bruin, 2023). In this way, AI reduces extraneous load while deliberately retaining the conceptual effort needed for understanding and transfer.

6.4 Practical implications

For SSWS, the framework offers a concrete, dyslexia-aware resource that can be adapted into workshops, study guides and one-to-one appointments. Table 3 can be shared with students as a quick reference that links common challenges to suggested AI use, human checkpoints and UDL principles.

For lecturers and programme leaders, the *AI Engagement Spectrum* provides a structured way to state expectations about AI in module handbooks and assessment briefs, helping to reduce ambiguity and inconsistency across a programme. Used at examination boards or curriculum review meetings, it can also surface where assessments rely heavily on product-only evidence and where additional process artefacts may be needed. As a worked example, Table 4 presents a constructively aligned “Engagement Spectrum” checklist that links intended learning outcomes, typical assessment tasks, required process evidence and AI permission levels, which lecturers can adapt directly into module handbooks and assignment briefs.

At institutional level, the Nested Scaffolding Model suggests how high-level AI and academic integrity policies, task-level decisions and students’ day-to-day study habits can be aligned. Rather than issuing generic warnings or permissions, universities can support departments in adopting locally appropriate AI policies that are transparent, inclusive and practically implementable.

6.5 Limitations and future work

This study has several limitations. The empirical strand involved a small number of specialist staff and a modest, self-selecting student sample from a single UK institution, so the findings cannot be assumed to represent all students with dyslexia or all disciplinary contexts. We have not yet carried out a full implementation study of the framework or measured impacts on attainment, workload or perceptions of academic integrity.

Future work will therefore focus on piloting the *AI Engagement Spectrum* and *AI Scaffold* in specific modules, collecting quantitative and qualitative data on usage patterns, learning outcomes, and staff and student experiences. Comparative work with other groups of neurodivergent learners, and with students without diagnosed learning differences, would help to test the generalisability of the Nested Scaffolding Model. Finally, as institutional AI policies and tools continue to evolve, there is a need for longitudinal research that tracks how permissions, practices and expectations around AI-supported study change over time.

7 Conclusion

This paper has presented the *AI Engagement Spectrum* and *AI Scaffold* as a dual-component, UDL-aligned system for integrating GenAI into the academic work of students with dyslexia. Developed from mixed-methods data with students and the SSWS, and grounded in constructive alignment, the framework offers a structured, tool-agnostic workflow that supports reading, writing and planning while embedding human-in-the-loop checkpoints. In doing so, it moves beyond individual tools to propose a transferable architecture for AI-supported assessment in higher education.

The framework's key contribution is to show how AI can be used to reduce extraneous cognitive load without erasing desirable difficulties, preserving students' authorship and critical engagement. By pairing an instructor-facing permission model with a student-facing workflow, the Nested Scaffolding approach gives institutions, lecturers and support services a shared language for setting AI expectations and designing accessible tasks. Rather than framing AI as either a threat or a panacea, the model illustrates how carefully bounded AI use can enhance autonomy and equity for students with dyslexia and profile-similar learners.

As an exploratory study with a small, context-specific sample, the work reported here is a starting point rather than a definitive solution. Looking ahead, future research can examine how the framework functions in larger and more diverse cohorts, and in disciplines beyond computing. It will also be important to investigate how sustained use of the *AI Scaffold* shapes students' confidence, self-advocacy and attainment, and how the *AI Engagement Spectrum* can be embedded within programme-level AI and assessment strategies. Such work would help institutions to translate rapidly evolving AI technologies into day-to-day practices that are both inclusive and pedagogically robust.

Appendix A Interview guide for SSWS

This appendix summarises the core prompts used in the semi-structured interviews with members of the SSWS. The interviewer used these questions flexibly, with follow-up probes as needed.

A.1 Requirements interviews with SSWS

1. Could you tell me about your role in the SSWS?
2. What kinds of students do you typically support in this role?
3. In your experience, what specific areas do students with dyslexia most frequently request additional assistance with?
4. Which types of academic tasks, assignments or assessments tend to be more challenging for students with dyslexia, and why?
5. What forms of academic support (for example study skills tuition, workshops or resources) are currently provided by the SSWS for these students?
6. What study-support or assistive tools do you currently recommend to students with dyslexia?

- (a) Which of these tools seem most popular or effective with students?
 - (b) What feedback, if any, have you received from students about these tools?
7. Does your team currently do anything to assess or monitor the impact of these tools and strategies on students' learning?
 8. To your knowledge, are students with dyslexia already using any AI-powered tools (for example ChatGPT, Grammarly, mind-mapping tools, Goblin Tools) in their academic work?
 - (a) If so, which tools and what do they tend to use them for?
 9. Are there any concerns or reservations within the SSWS or the wider university about incorporating AI tools alongside the support that is already available?
 10. Thinking specifically about academic integrity, what kind of conditions or guidelines would AI-powered tools need to satisfy in order to be acceptable in higher education?
 11. If such guidelines existed, how do you think they should be implemented and communicated in practice?
 - (a) For example, should this be more tutor-led, student-led or embedded in module documentation and assessment briefs?
 12. Are you worried about risks such as over-reliance on AI, skill atrophy, or AI tools producing biased or inaccurate outputs?
 - (a) How might these risks affect your work with students?
 13. From your perspective, what features or functions would you like an AI-powered tool or workflow to have in order to support students with dyslexia more effectively?
 14. How do you think student feedback on any new AI-based tools or workflows should be collected and used?
 15. Is there anything else you would like to add about the opportunities or risks of AI in supporting students with dyslexia?

A.2 Workflow feedback interview with SSWS expert

The following prompts were used to gather feedback on the prototype *AI Scaffold* workflow from an experienced SSWS practitioner.

1. What was your overall first impression of the *AI Scaffold* workflow?
2. Did the overall structure of the workflow (sections and steps) make sense to you?
3. Were any parts of the workflow or the prompts confusing or unclear?
4. How do you feel about the wording and tone of the workflow (including the student-facing prompts and explanations)?
 - (a) Do you think the language is appropriate for your students?

5. Do the categories we use (for example reading and summarising, planning and structuring, drafting, proofreading) reflect how you actually work with students in your study-skills appointments?
6. When you look at the steps in order, does the sequence feel logical and practical for students to follow?
7. How would you imagine using this workflow in your own practice?
 - (a) For example, would you use it during one-to-one appointments, workshops, or as a handout for students to use between sessions?
8. Do you think students would be able to follow these steps independently between appointments?
9. Are there any steps or elements you would add, remove or re-order to make the workflow more useful or realistic?
10. Does the way AI use is described in the workflow match your expectations about ethical use and academic integrity?
11. Do you see any risks, misunderstandings or unintended consequences that we should address before using this workflow with students?
12. Is there anything else you would like to change or suggest to improve the *AI Scaffold* for your context?

Appendix B Student survey instrument

This appendix summarises the questionnaire “*Use of AI-powered tools in higher education*” used in this study. Items were grouped into sections focusing on study challenges, tool use, perceptions of AI and ethical concerns. The questionnaire combined open-ended prompts with structured response options. The items below list the items used to produce the descriptive statistics reported in Section 4.

Consent

1. I have read the information about the study and consent to participate.

Background and study context

2. What is your field of study?

Study challenges and current strategies

3. What are the biggest challenges you face when planning and organising your academic work? (response options)

4. What tools or strategies do you currently use for brainstorming and structuring your ideas? (response options)
5. How do you manage large reading and writing tasks? Are there any specific techniques that help? (response options)
6. How helpful do you find traditional note-taking and outlining methods (for example paper-based notes, flashcards, etc.)? (response options and open-ended)
7. Why do you find traditional note-taking methods unhelpful? (response options and open-ended) (*Displayed only if respondents indicated low helpfulness in Q6.*)
8. How well do your current strategies support your academic success? Do you find gaps in the support? (response options and open-ended)

Use of institutional and other digital tools

9. Do you use any of the tools provided by the SSWS? If yes, which ones and what do you use them for? (response options and open-ended)
10. Have you used any digital tools, not provided by the SSWS, to support your learning? If yes, please specify which ones. (response options and open-ended)

Use of AI-powered tools and desired features

11. Do you find AI-powered tools (such as ChatGPT or Grammarly) useful in your academic work? If yes, in what ways? (response options and open-ended)
12. Do you think AI-powered mind-mapping tools could help with brainstorming and organising ideas? Why or why not? (response options and open-ended)
13. What features would you look for in an AI-powered tool to help you with brainstorming and idea creation? (response options)

Ethical concerns and guidance

14. Are you concerned about AI tools affecting your academic integrity? If so, what concerns do you have? (response options and open-ended)
15. Do you feel that the university has provided guidance on how to use AI ethically in your studies? (response options)
16. What do you think would prevent students from declaring the use of AI in their work? (response options)

Appendix C Detailed mapping of the *AI Scaffold* and engagement spectrum to UDL 3.0 guidelines

Table 5 UDL Aligned AI Workflow for Supporting students with dyslexia

Dyslexia-related Challenge	<i>AI Scaffold</i> action	UDL Principle	UDL Guideline & Checkpoint (v3.0)	Justification / Elaboration
Slow, laborious decoding of dense text; reading fatigue.	Use TTS to read text aloud.	Representation	Guideline 1: Perception Checkpoint 1.2: Support multiple ways to perceive information (e.g., auditory information).	- Offers an auditory alternative to text. - Bypasses the phonological decoding barrier. - Reduces cognitive load, freeing resources for comprehension.
High cognitive load from dense text; difficulty extracting key points.	Use AI to summarise long articles into key points.	Representation	Guideline 3: Building Knowledge Checkpoint 3.2: Highlight and explore patterns, critical features, big ideas, and relationships.	- Condenses text to reduce the cognitive load. - Highlights critical features and big ideas. - Lets learners focus on analysis, not decoding.
Limited academic vocabulary; difficulty with formal academic English and writing style.	Use AI to explain grammatical errors or suggest alternative phrasing.	Representation	Guideline 2: Language & Symbols Checkpoint 2.1: Clarify vocabulary, symbols, and language structures.	- Clarifies complex vocabulary and language on demand. - Explains the conventions of academic discourse. - Develops academic vocabulary and writing style.
Difficulty overcoming “writer’s block”; challenges with initial idea generation.	Use AI to brainstorm topics or ideas for an assignment.	Engagement	Guideline 7: Welcoming Interests & Identities Checkpoint 7.1: Optimise choice and autonomy.	- Lowers the barrier to starting (an executive function scaffold). - Optimises choice by providing a range of starting points. - Reduces blank page’ anxiety while preserving student agency.
Difficulty with the mechanics of spelling	Use speech-to-text to dictate initial	Action & Expression	Guideline 4: Interaction Checkpoint 4.1: Vary	- Bypasses mechanical barriers of typing and spelling. - Captures thoughts at the speed of speech.

Table 5 (continued)

Dyslexia-related Challenge	AI Scaffold action	UDL Principle	UDL Guideline & Checkpoint (v3.0)	Justification / Elaboration
and slow typing speed.	draft text.		and honour the methods for response, navigation, and movement.	- Separates composition from the physical act of writing.
Difficulty with formal academic expression; slow writing speed.	Use AI to expand key points from an outline into draft paragraphs.	Action & Expression	Guideline 5: Expression & Communication Checkpoint 5.2: Use multiple tools for construction, composition, and creativity.	- Acts as a tool for construction, scaffolding the drafting process. - Frees students to focus on higher-order logic and argument. - Requires revision to ensure ownership and academic voice.
Difficulty with linear organisation and structuring arguments.	Use AI to create a potential outline from notes or a research question.	Action & Expression	Guideline 6: Strategy Development Checkpoint 6.3: Organise information and resources.	- Provides a modifiable structure for planning and organisation. - Supports the executive function of organising information. - Translates non-linear thinking into a linear format.
Feeling overwhelmed; difficulty breaking down large assignments; procrastination.	Following a step-by-step workflow with clear stages and checkpoints.	Engagement	Guideline 8: Sustaining Effort & Persistence Checkpoint 8.2: Optimise challenge and support.	- Breaks large tasks into manageable sub-tasks. - Balances challenge and support to reduce cognitive overload. - Sustains effort by clarifying the path forward.
Tendency to accept information uncritically; risk of skill atrophy or AI “hallucinations”.	The workflow requires checkpoints for accuracy verification and personalisation.	Engagement	Guideline 9: Emotional Capacity Checkpoint 9.3: Promote individual and collective reflection.	- Prompts critical evaluation and metacognitive reflection. - Requires active verification of AI output to counter passivity. - Builds AI literacy and self-regulation skills.

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Declarations

Competing interests The authors have no competing interests to declare that are relevant to the content of this article.

Ethics approval and consent to participate Ethical approval for this study was obtained through University's online ethics application system (LEON). All participants received an information sheet outlining the study's purpose, procedures, and their rights. Informed consent was obtained from all participants prior to data collection. No data were collected from individuals who did not provide consent.

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